



PRODUCT NAME

Professional Toothpaste Formula

Description

Fluoride toothpaste with sorbitol, silica, fluoride, flavor, and preservative.

Technical Specifications

pH Level:	7
Viscosity:	Medium
Shelf Life:	24 months
Batch Size:	100-500 kg
Processing Time:	2-4 hours
Temperature:	Room temperature
Storage Conditions:	Cool, dry place
Equipment:	Standard mixer
Certification:	Meets industry standards

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Purified Water	Aqua	NaN%	Solvent
2	Sorbitol Solution	Sorbitol	NaN%	Humectant
3	Hydrated Silica	Hydrated Silica	NaN%	Abrasive
4	Glycerin	Glycerin	NaN%	Humectant
5	Sodium Lauryl Sulfate	Sodium Lauryl Sulfate	NaN%	Foaming Agent
6	Sodium Monofluorophosphate	Sodium Monofluorophosphate	NaN%	Active Ingredient (Fluoride)
7	Flavor	Aroma	NaN%	Flavoring Agent
8	Sodium Saccharin	Sodium Saccharin	NaN%	Sweetener
9	Xanthan Gum	Xanthan Gum	NaN%	Thickening Agent
10	Sodium Benzoate	Sodium Benzoate	NaN%	Preservative
11	Titanium Dioxide	Titanium Dioxide	NaN%	Colorant
12	Sodium Hydroxide	Sodium Hydroxide	NaN%	pH Adjuster
Total			100%	

Manufacturing Process

Phase 1: Manufacturing Phase

Phase 2: Manufacturing Phase

Phase 3: Manufacturing Phase

Phase 4: Manufacturing Phase

Phase 5: Manufacturing Phase

Phase 6: Manufacturing Phase

Phase 7: Manufacturing Phase

Phase 8: Manufacturing Phase

Phase 9: Manufacturing Phase

Required Equipment

Standard mixer

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Cool, dry place

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Meets industry standards

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 100-500 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply as needed according to product instructions